

of designs available:

Basics. Resistors, capacitors, inductors. This is simply to appreciate the role of these components in any circuit. CircuitMod also includes voltage dividers for a set of voltage-generation, Thevenin's and Norton's theorems.

AC circuits. Circuits depicting the behaviour of capacitors and inductors under an alternating current (AC) source and to different frequencies can easily be studied with the circuit spread under this section. Concepts of impedance, series resonance and parallel resonance are not difficult to grasp anymore.

Passive circuits. Everything from filter and crossover circuits, to transformers and differentiators are illustrated here. There are also circuits that help you understand how exactly three- and four-way light switches work, why wheatstone bridge balancing is important to avoid current flow between its legs, and circuits performing inductive kickback action.

Power factor analysis. Let us take an example to understand why this style of learning is so powerful. Consider an inductor being driven by AC voltage. Below is the explanation for the circuit as taken from Falstad website:

"The colours indicate power consumption; red means that a

component is consuming power and green means that the component is contributing power. The left side of the circuit represents the power company's side and the right side represents a factory (with a large induction motor). The highly-inductive load is causing the power company to work a lot harder than normal for a given amount of power delivered. The graph on the left indicates the power lost in the power company's equipment (the resistor at top left). The graph in the middle is the power delivered to the factory. The graph on the right is the power delivered to the inductor (and then returned, causing the time average of power delivered to be zero). Even though a peak power of 40mW

is being delivered to the factory, 200mW is being dissipated in the power company's wires. This is why power companies charge extra for inductive loads."

The above clarifies why thorough analysis of even simple circuits is very important and how exactly CircuitMod chips in here.

A little more. Apart from the above, there are circuits involving diodes—be it rectifier or diode characteristics, voltage multipliers, communication-related circuits like detectors, transistors and op-amps in their various configurations, oscillators, JFETs, MOSFETs, timers, logic families, digital circuits, phase-locked loops and transmission lines; basically, the entire package.

Tried and tested

The tool is at your disposal. How effective is your learning depends on how much you exploit the available resources. Play around with the options. You can not only modify the circuit, but also the way these look. Replace resistors with European-style ones, change the screen background to white, reverse conventional current direction—the options are quite a handful. Explore and enjoy! **EFY**

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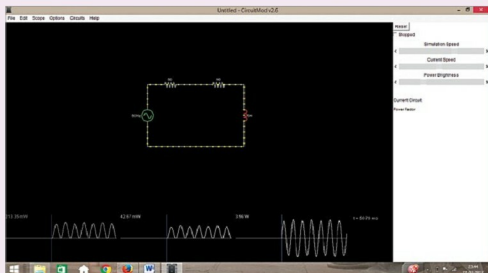


Fig. 3: Power factor analysis using CircuitMod

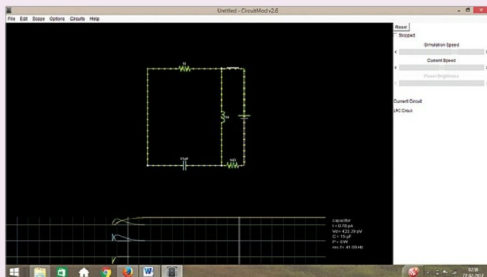


Fig. 4: The same LRC circuit with stacked traces and showing specific details of the highlighted capacitor